

Characterization approaches for fiber-reinforced composites

Simulation and Design of Load-Displacement Response of Hybrid FRC Flexural Elements

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Presentation to *fib* Task Group 4.1

Areas of Need

- Computational tools to obtain the material properties from standard Flexural test
- Design of Hybrid FRC and how these ideas challenge our 100-year-old approach to the stress block diagram
- Discuss the variations in the test results and how to overcome them. (We need to stop penalizing FRC Materials and accept Variability as an inherent integral part of the material)
- Software tools for the design address the need for expedited design and evaluation
- Main Philosophical questions:
 - What is the efficiency of concrete in tension?
 - How low is low?
 - What are we doing to change the narrative?

How do we approach the modeling problem?

- Integrate the three essential components of the undergraduate level “Mechanics of Materials” approach to the 1-D flexural response of composite systems:

- 1- Equilibrium equations, in terms of statics, internal and external equilibrium of forces and moments
- 2- Constitutive relationships, multi-linear tensile and compressive 1-D elastic-hardening/softening response of matrix and reinforcement (various models available)
- 3- Compatibility conditions, Strain-curvature compatibility approach

Seek closed-form solutions to a variety of combinations

The Stress-Strain-Compatibility Relationship, the Limit States

The Stationary and Moving Control Points

Formulations use only two variables of Maximum Tensile Strain, β and the location of the Neutral Axis, k , and input of material and geometrical properties

$$\varepsilon_{\max} = \beta \varepsilon_{cr}$$

$$\varepsilon(y) = \beta'(y) \varepsilon_{cr} \quad \text{where} \quad \beta' = \frac{\varepsilon(y)}{\varepsilon_{cr}} = \frac{\beta y}{(1-k)h} \quad \lambda = \frac{-\beta k}{1-k}$$

$$\chi' = \frac{-(\alpha + k - 1)\beta}{1-k} \quad \chi = \frac{(\alpha - k)\beta}{1-k}$$

$$\varepsilon_{\max} = \beta \varepsilon$$

Normalized
Strain

Derivation of Moment-Curvature Relationship, two variables: Max. tensile Strain β , and location of N.A.

Increment max strain, β $0 \leq \varepsilon_{\max} \leq \varepsilon_{tu}$



Strain Distribution
Stress Distribution



$\Sigma F = 0$, determine the
location of N.A. k



Force efficiency
factors



Moment:
 $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$
Curvature: $\phi = \varepsilon_c / kh$

Strain distribution $\varepsilon(y) = \beta'(y) \varepsilon_{cr}$ where $\beta' = \frac{\varepsilon(y)}{\varepsilon_{cr}} = \frac{\beta y}{(1-k)h}$

Force Equilibrium*

$$F(\beta) = b \int_0^h \sigma(\beta') dy = F_{CT} + F_{CC} + F_{ST} + F_{SC} = 0$$

Moment

$$M(\beta) = b \int_0^h \sigma(\beta') y dy \quad M = m' M_{cr} \quad M_{cr} = \frac{1}{6} b h^2 E \varepsilon_{cr}$$

Curvature

$$\phi = \frac{\beta \varepsilon_{cr}}{(1-k)h} \quad \phi = \phi' \phi_{cr} \quad \phi_{cr} = \frac{2 \varepsilon_{cr}}{h}$$

Force efficiency Factor*

Tension domain $\eta_{ST}(\beta) = \frac{F_{ST}}{F_{CT} + F_{ST}}$

$$\eta_{CT}(\beta) = \frac{F_{ST}}{F_{CT} + F_{ST}}$$

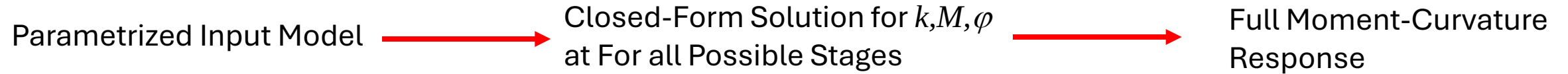
Compression domain $\eta_{SC}(\beta) = \frac{F_{SC}}{F_{CC} + F_{SC}}$

$$\eta_{CC}(\beta) = \frac{F_{CC}}{F_{CC} + F_{SC}}$$

* While $\Sigma F = 0$, still monitor the individual forces

Flexural Modeling of Reinforced FRC/UHPC (Hybrid Approach)

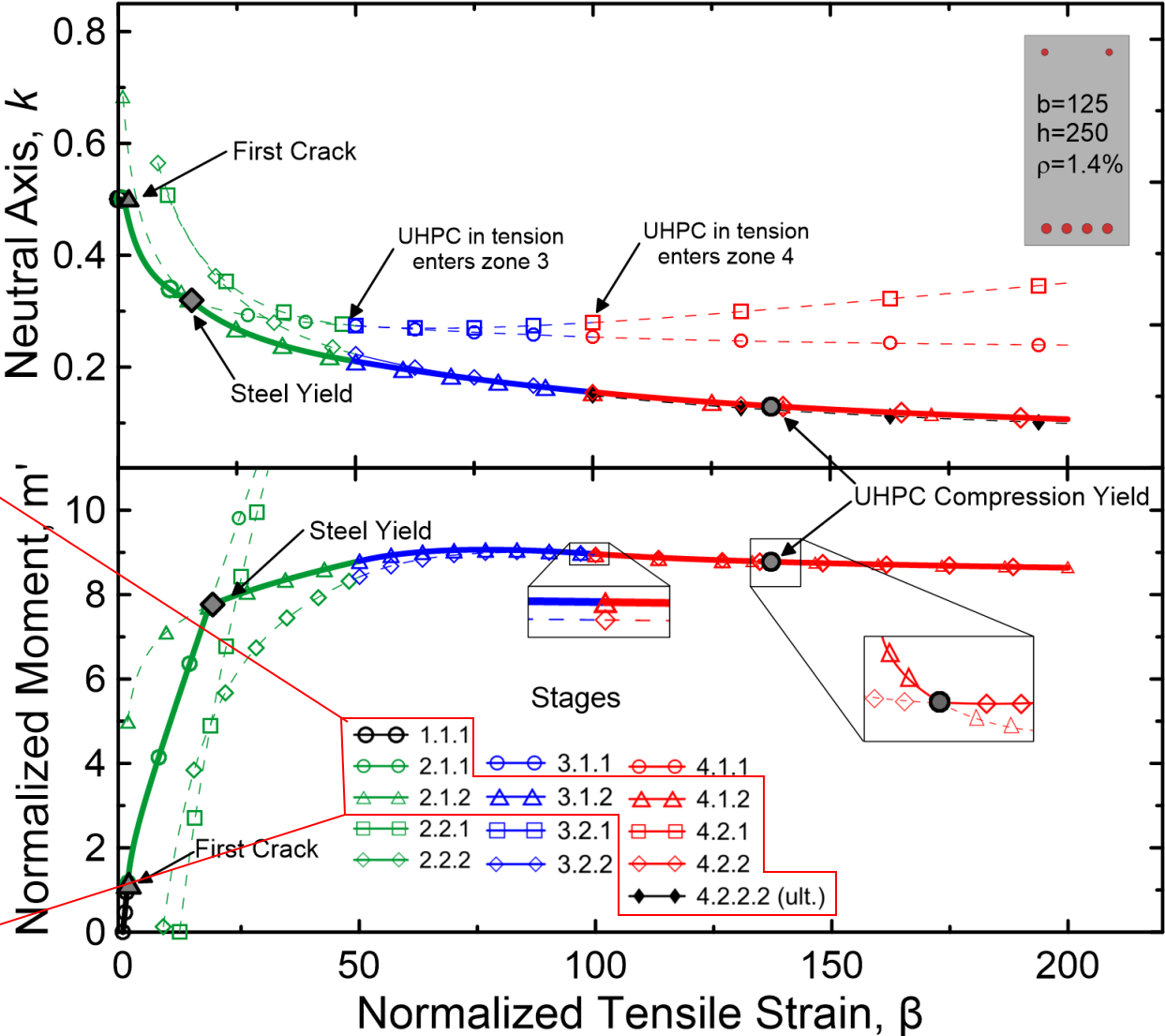
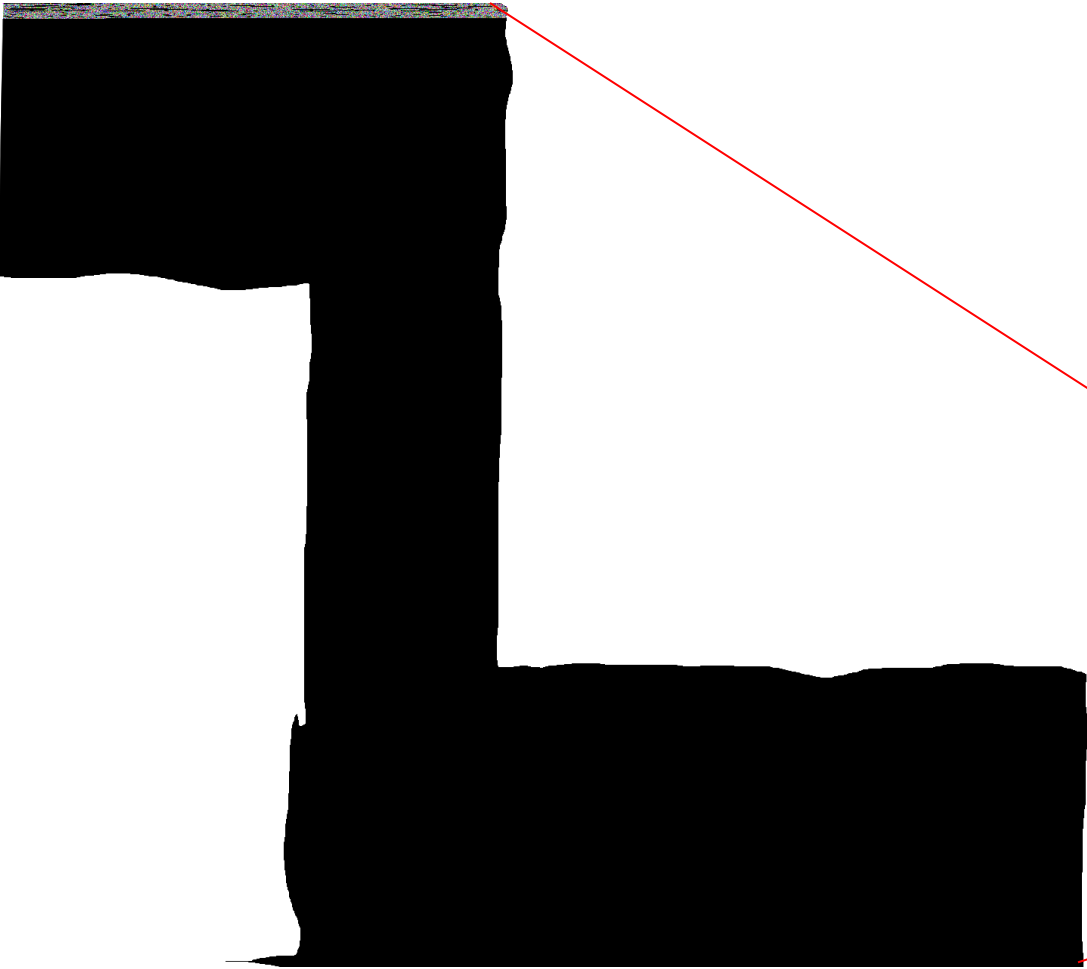
- Closed-form based analytical Model for flexure simulation and Inverse analysis



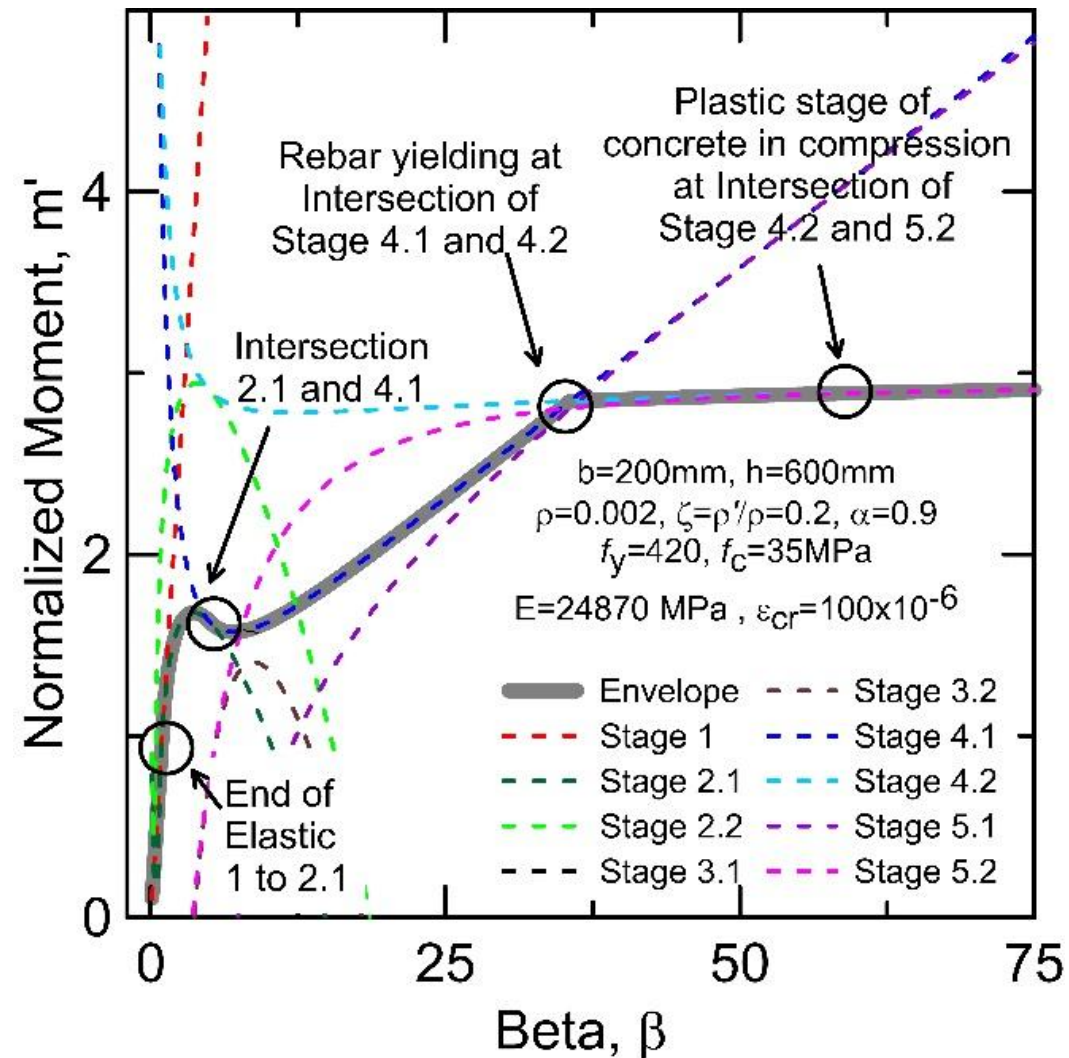
Bonus: Limit States

Interaction of the Stages, limit States, and the coordinates of Limit states and ranges.

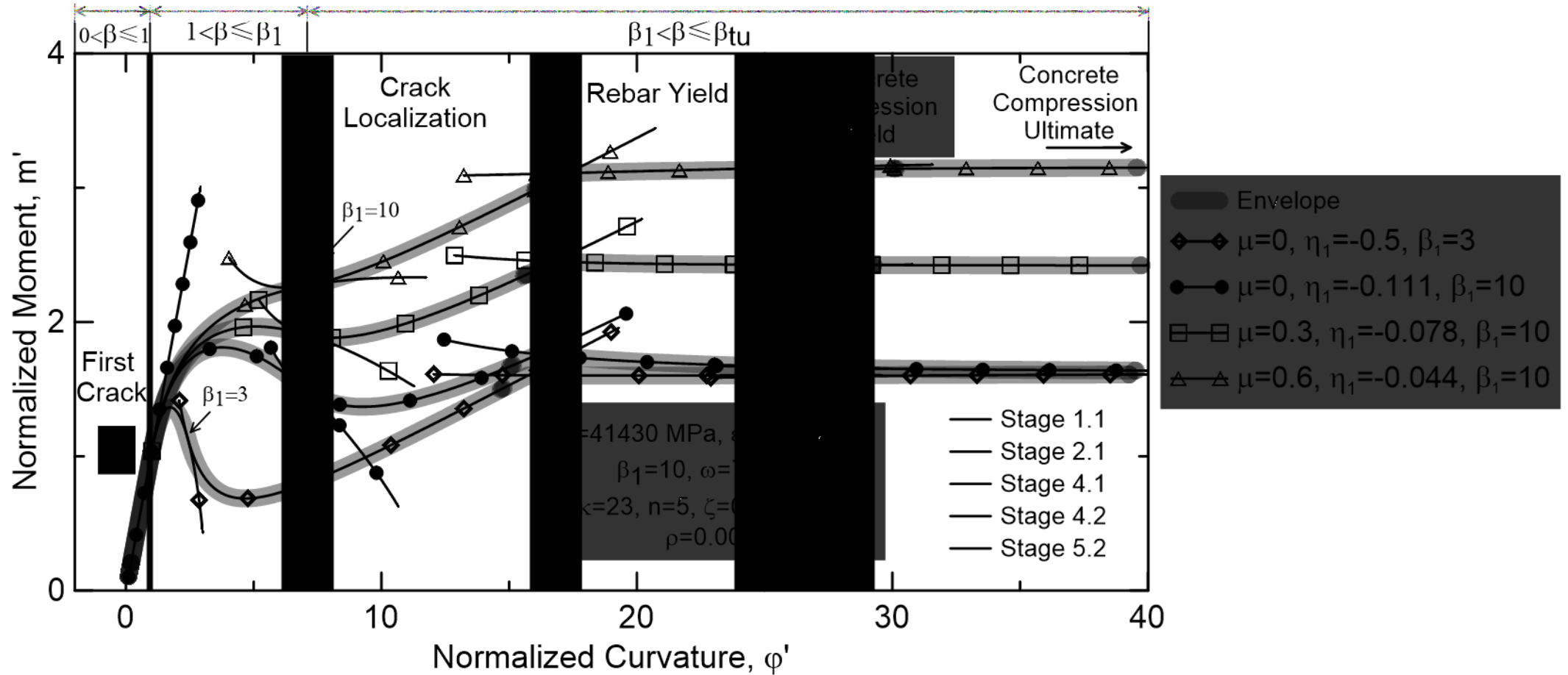
Envelope Formation for all possible stages



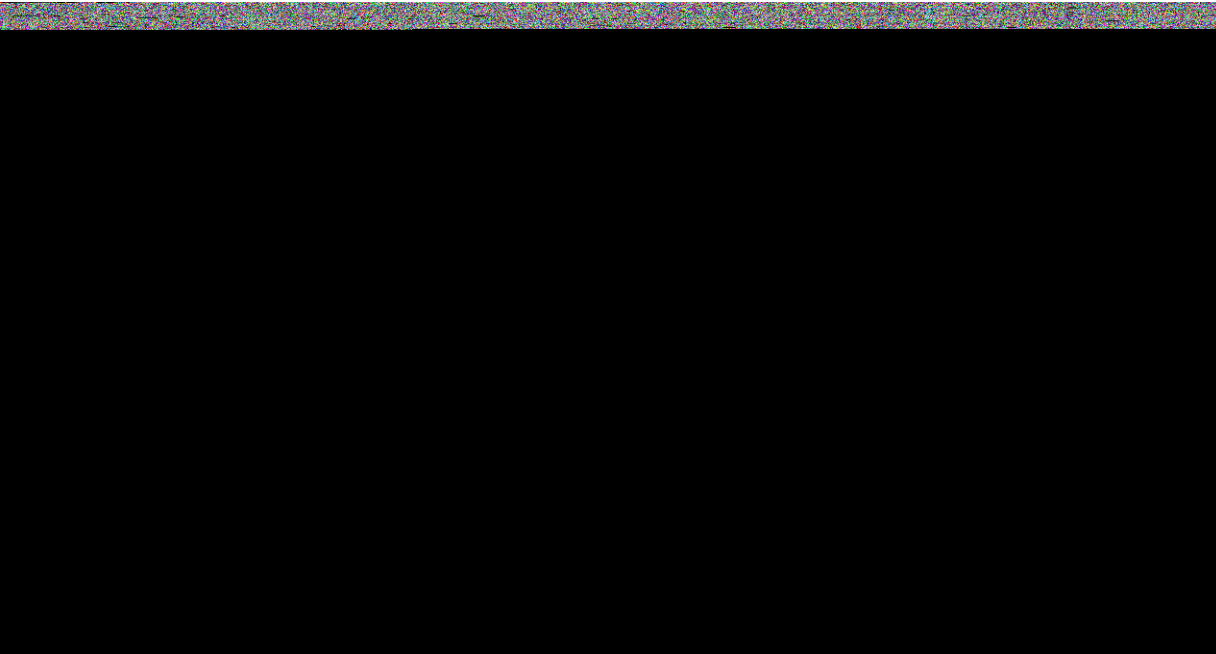
Generating moment by imposing normalized maximum tension strain, β



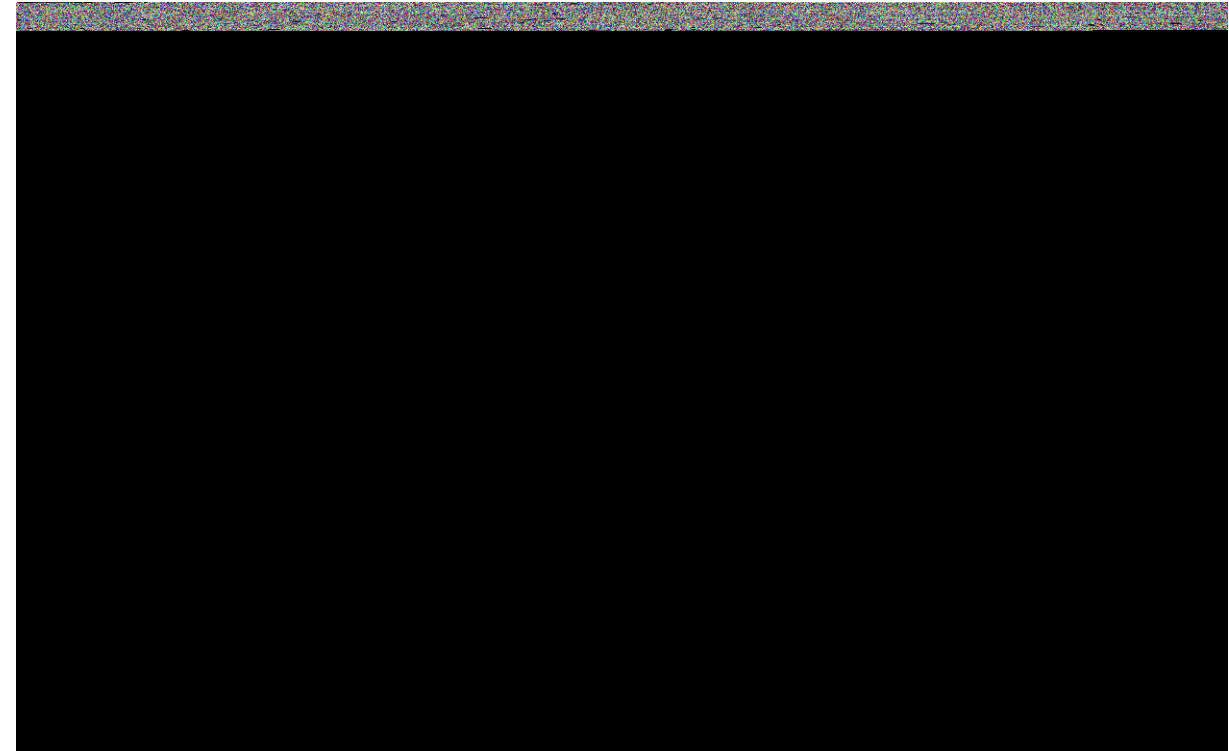
The intersection points of Moment-Curvature RC ($\mu=0$) and HRC ($\mu=0.3-0.6$), Get Limit states



Excel Versions of modeling back-calculation and Hybrid RC design, Versions: 2T2C, 2T2C2S, 3T2C,3T2C2S, 4T,2C2S_SH, 2T2C2S_PM



- Quadrilinear Tension- Bilinear compression for 3PB and 4PB
- PM Diagram for columns

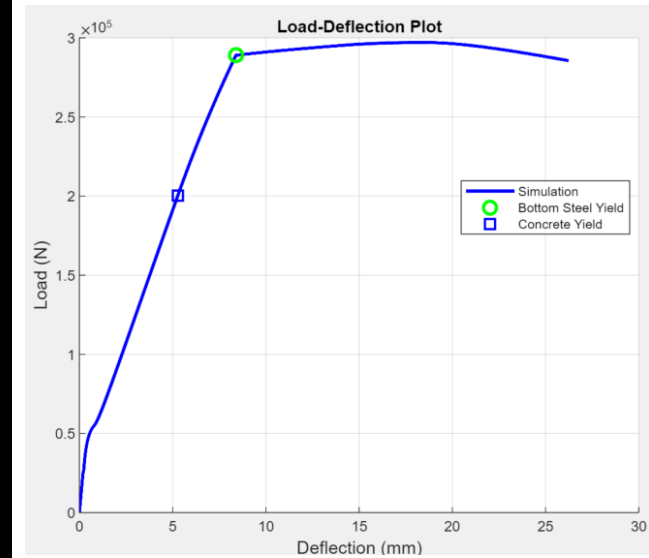


- Bilinear Tension- Bilinear comp. Bilinear Reinforcement for 3PB and 4PB
- Quadrilinear Tension- Bilinear hardening/softening comp. and Reinforcement for 3PB and 4PB

Applicability to interactive design software

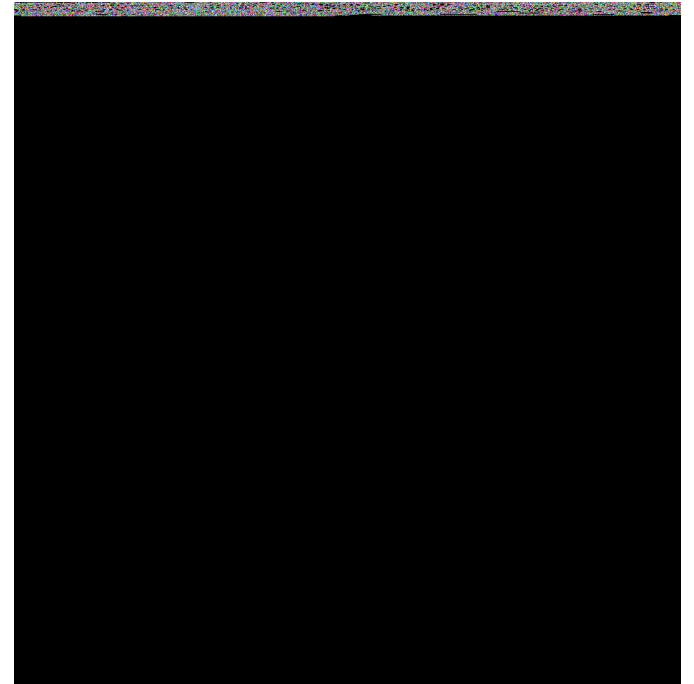
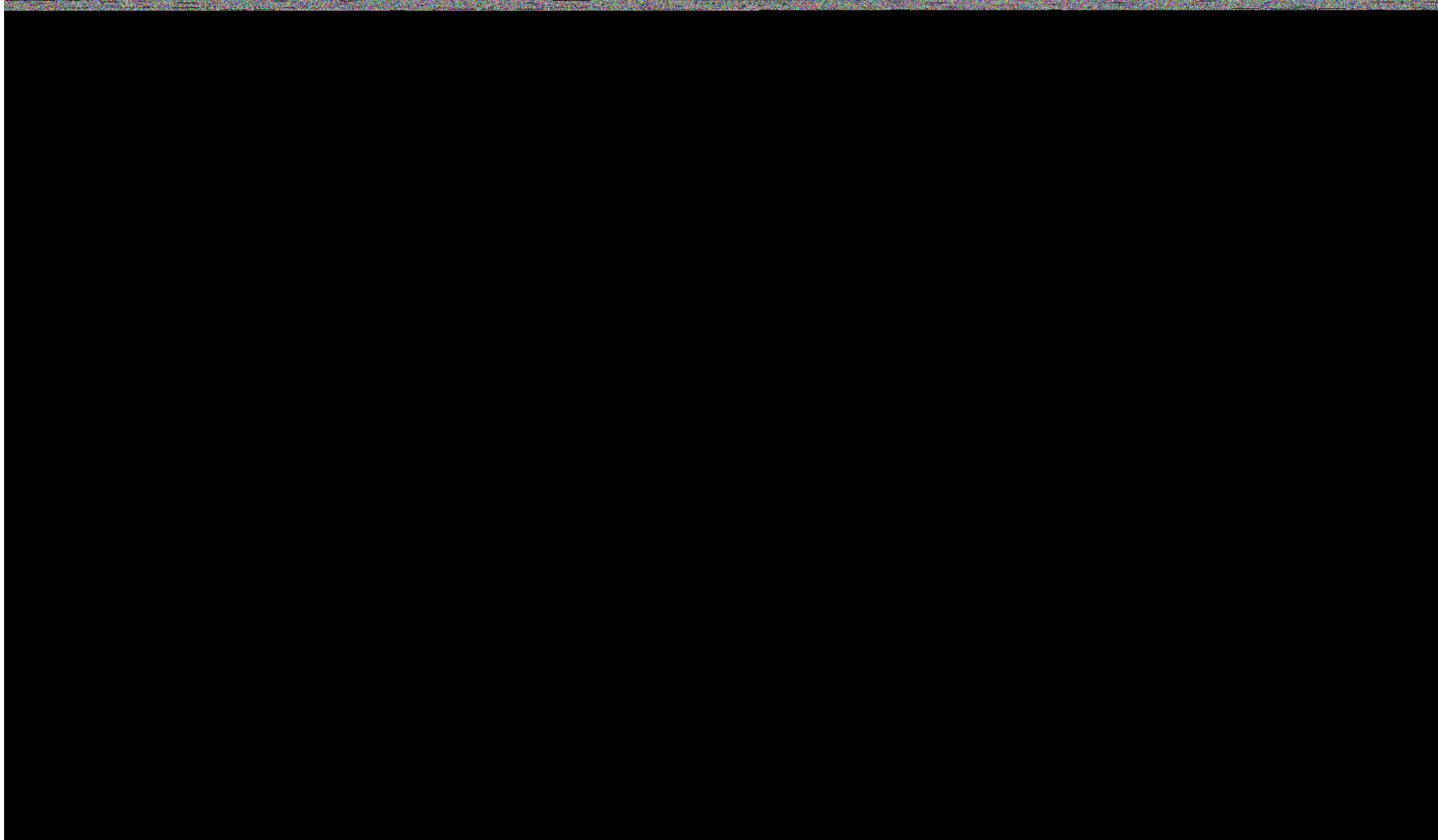


Validation of closed-form approach for Reinforced Hybrid Members



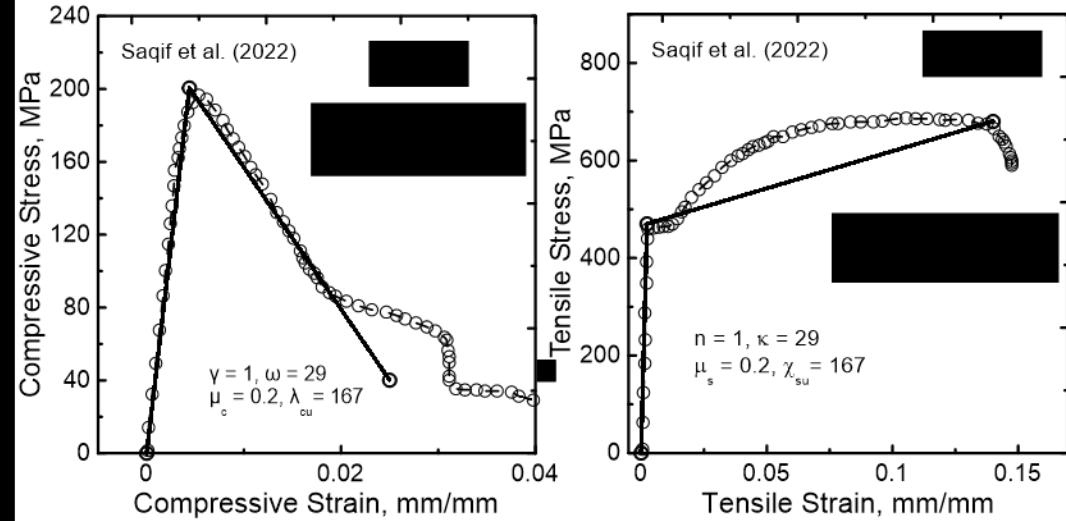
Inverse Analysis for characterization of FRC and UHPC (Min four Tensile segments)

-Quad-linear model



Validation of closed-form approach for Reinforced UHPC Members:

Material Characterization:

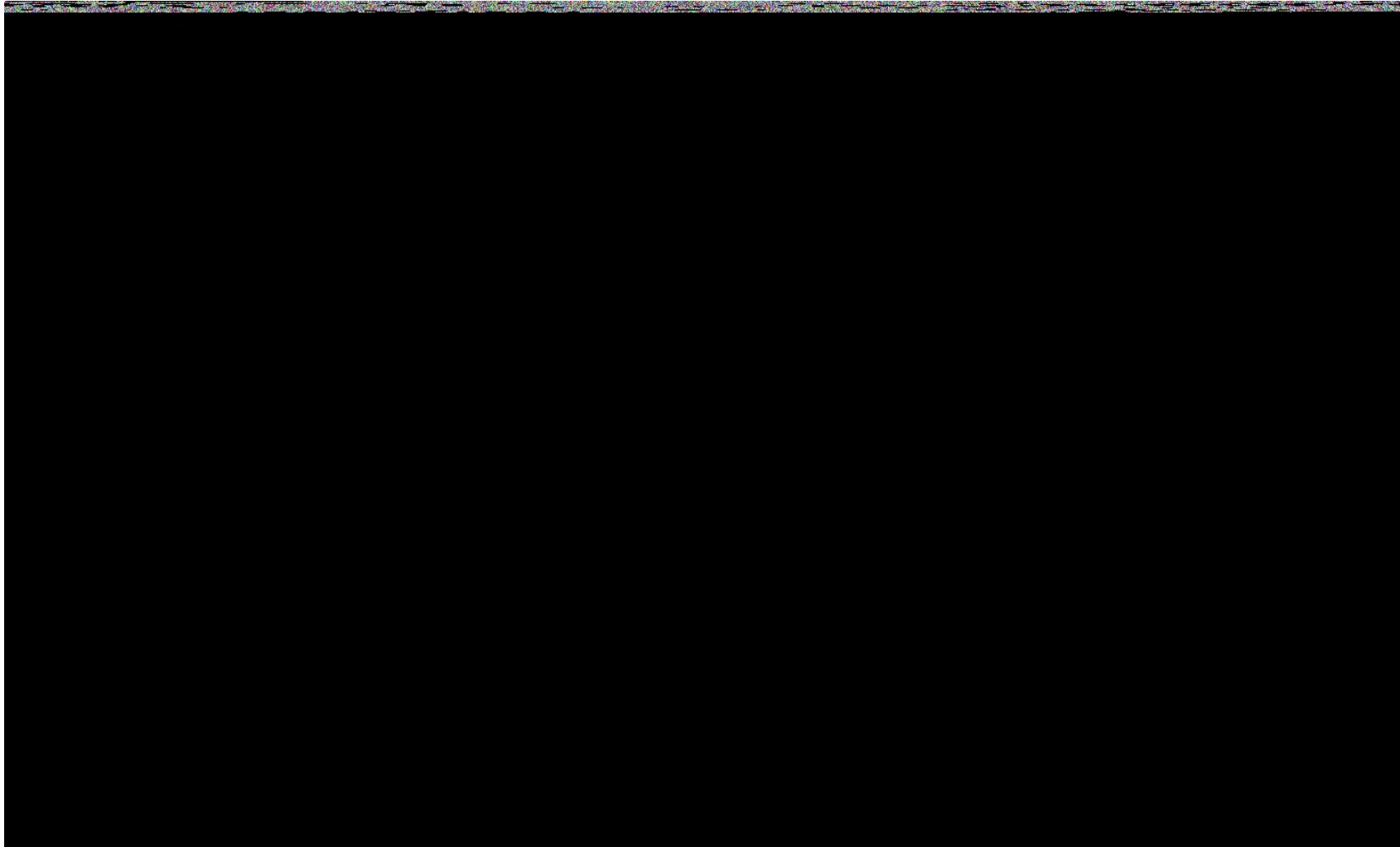


Validation:

Stress
Distribution at
Limit States

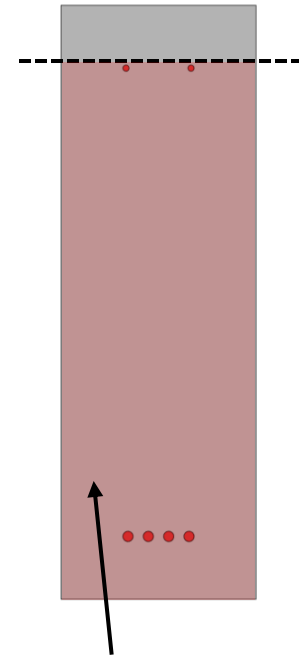
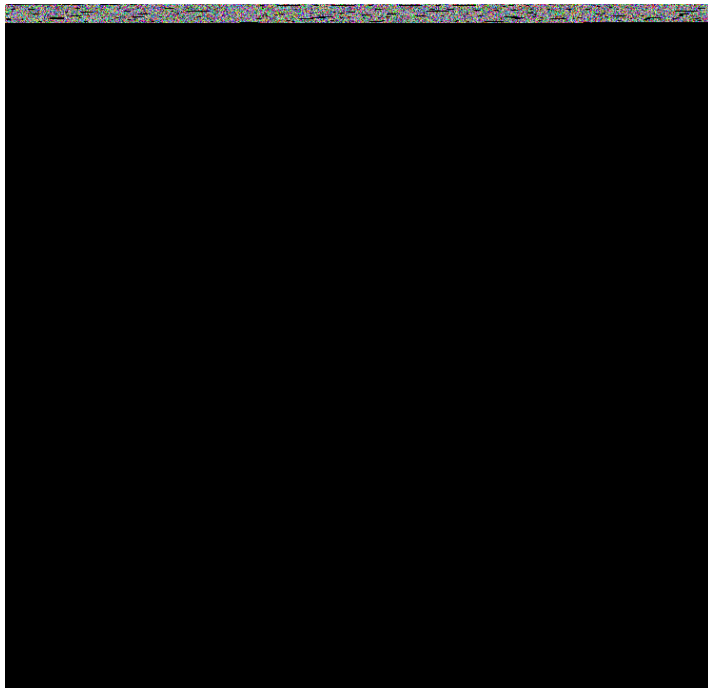
How do we justify the efficiency of concrete with and without fibers in conventional applications?

- Closed-form solution of force efficiency in terms of material properties
- Dimensionless design chart for FRC/HRC/UHPC

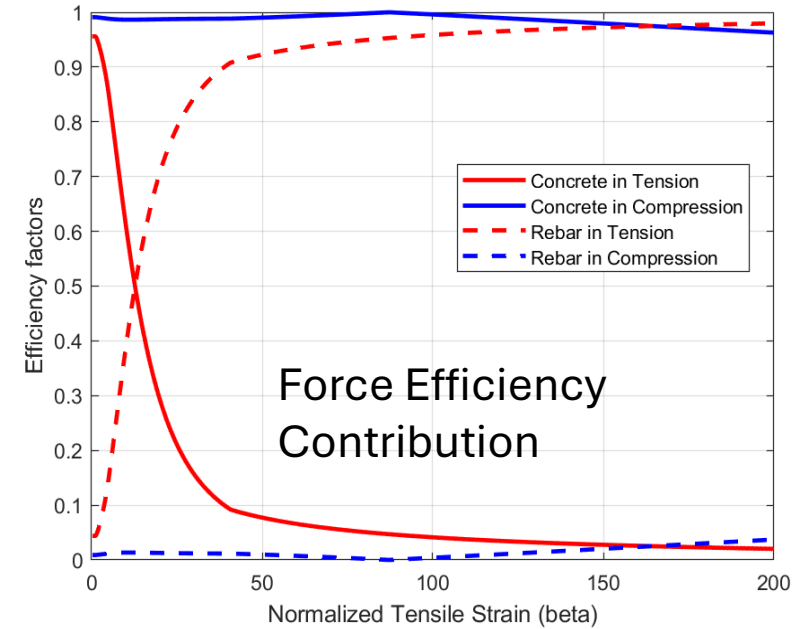
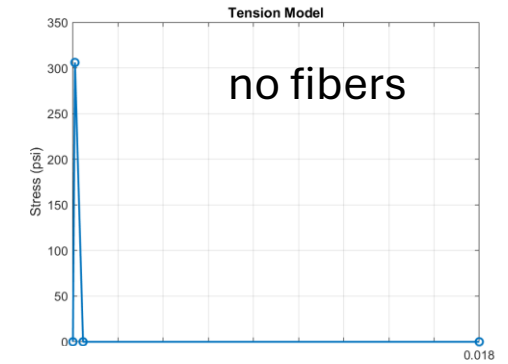
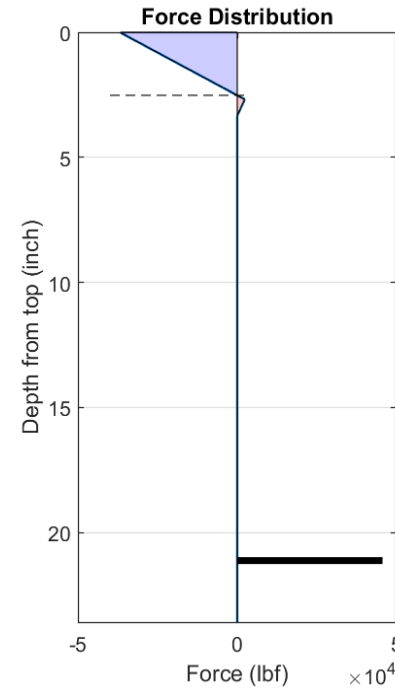


Design of hybrid (Effect of fibers on Materials Efficiency):

Case 1- Plain RC, No fibers



No contribution
from concrete in
Tension.



$$\rho = 0.28\%$$

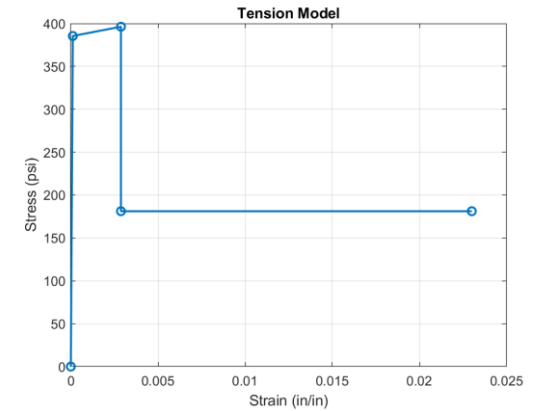
$$\mu = 0 \quad \text{No residual strength}$$

$$\text{Size} = 190 \text{ mm} \times 600 \text{ mm}$$

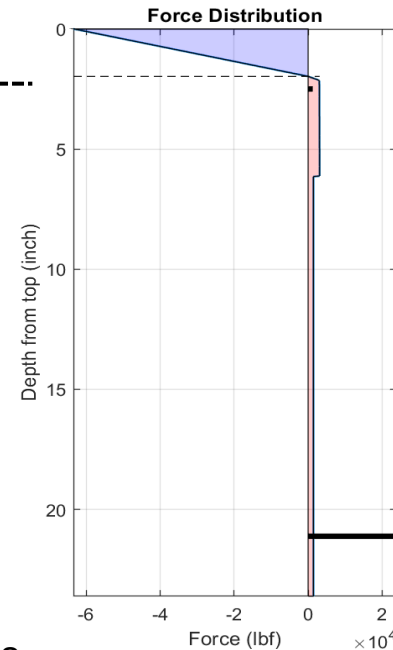
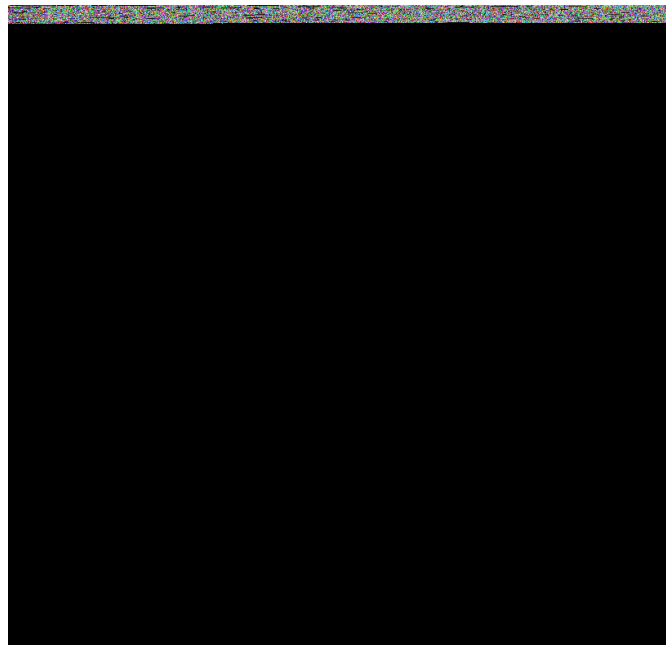
Design of hybrid (Effect of fibers on Materials Efficiency):

Case 2- Hybrid RC, Steel fibers, Min Reinforcement

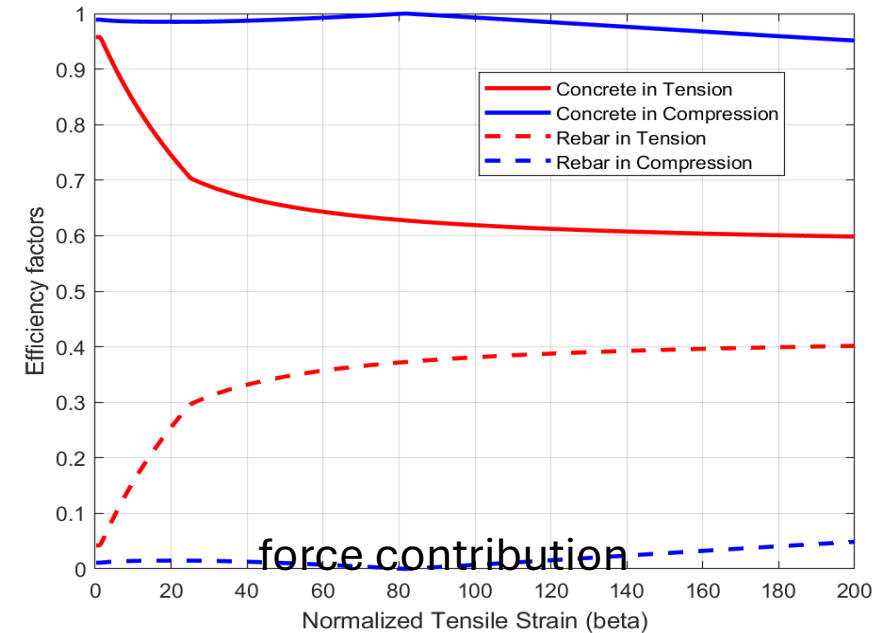
$$\rho = 0.13\%$$



$\mu = 0.5$ residual strength (less than 200 psi)



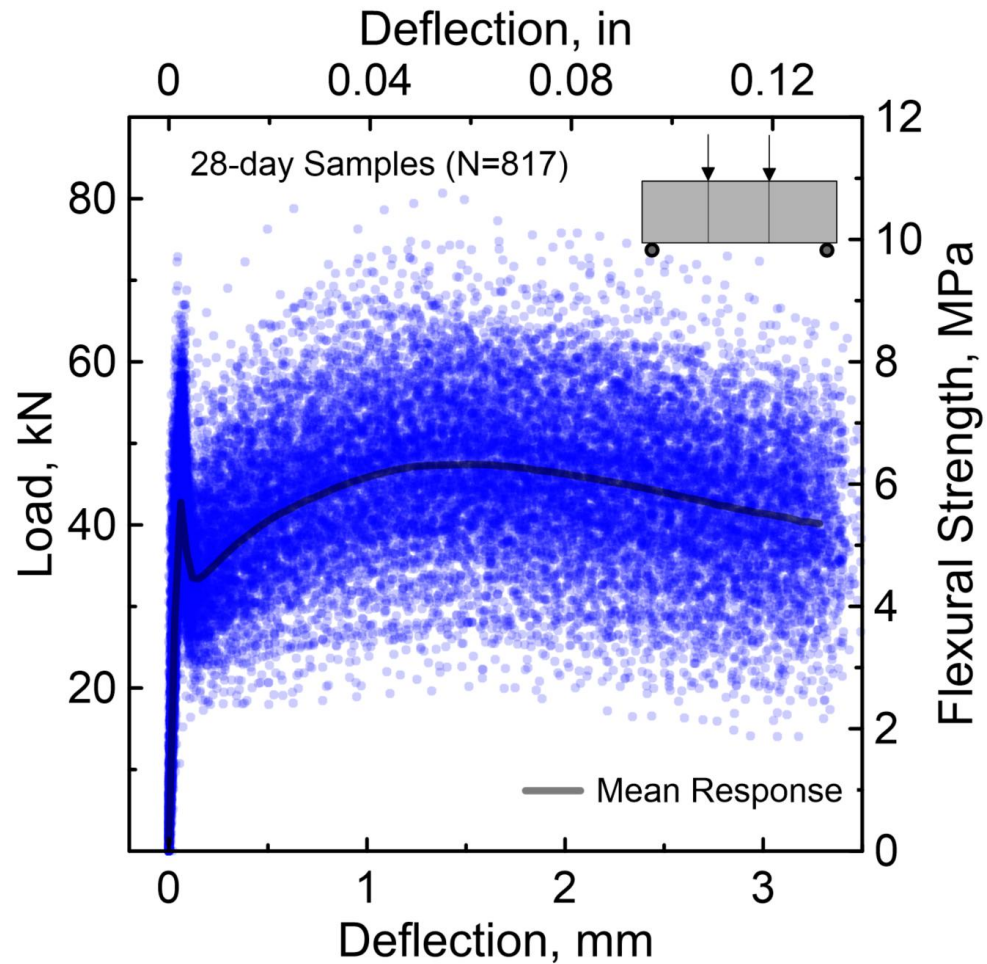
Concrete in Tension starts
carrying load immediately
after cracking



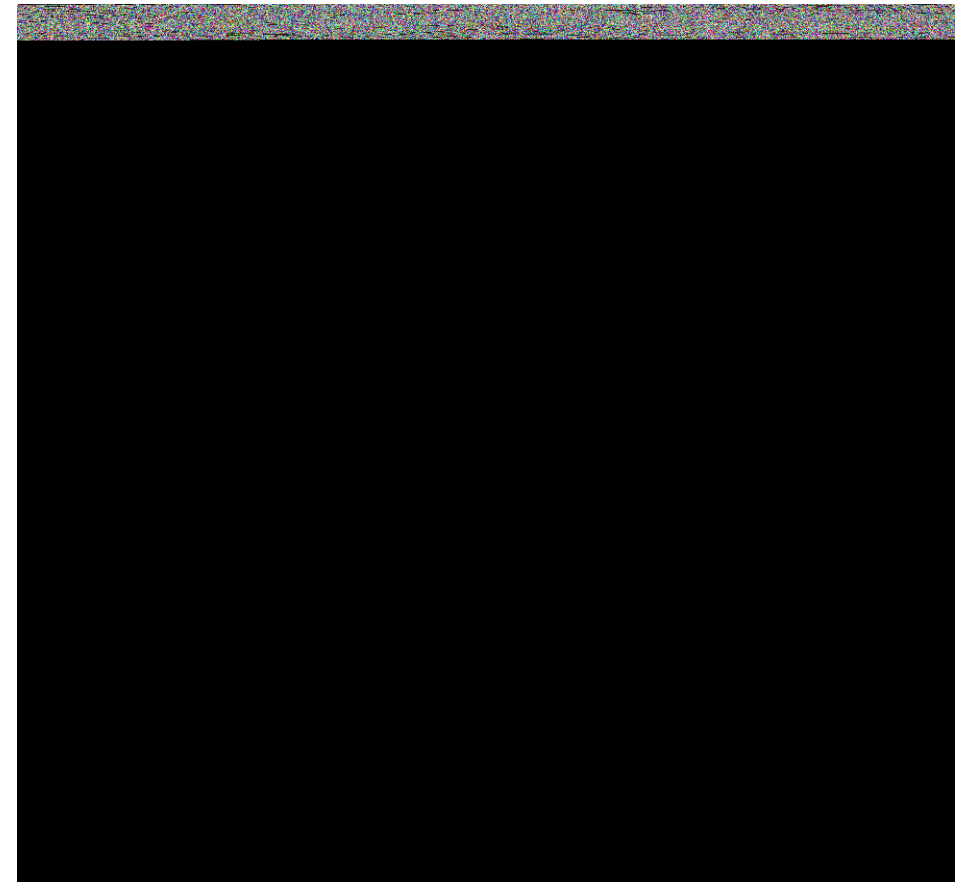
$$\rho = 0.13\%$$

Size = 190 mm x 600 mm

What is the utility of the modulus of rupture as an indicator of performance of fibers?

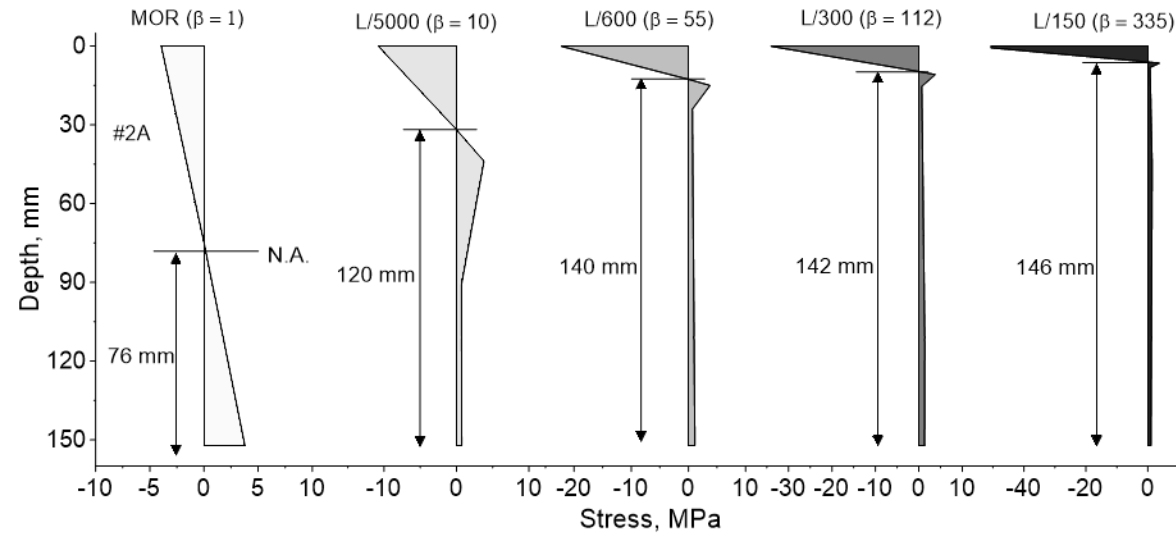


No influence of FRC on First Cracking Moment

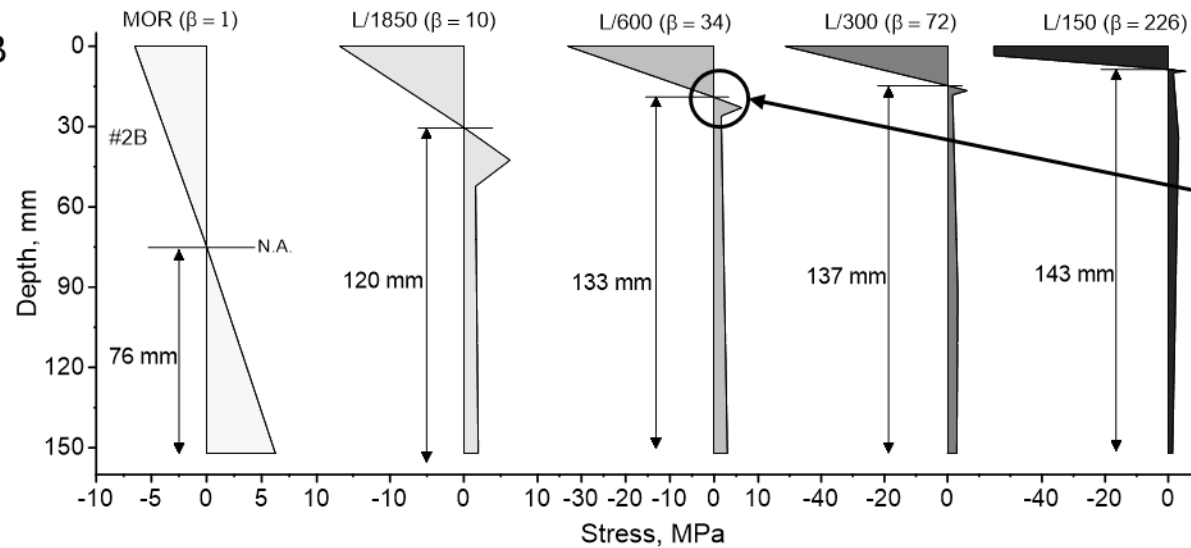


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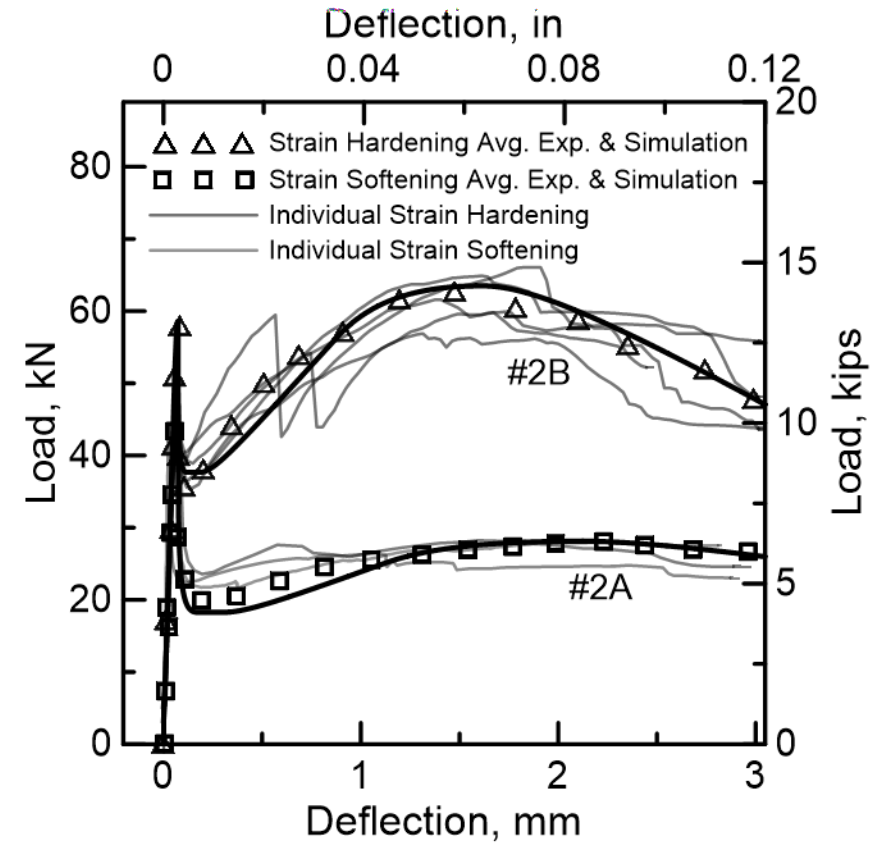
Set 2A



Set 2B

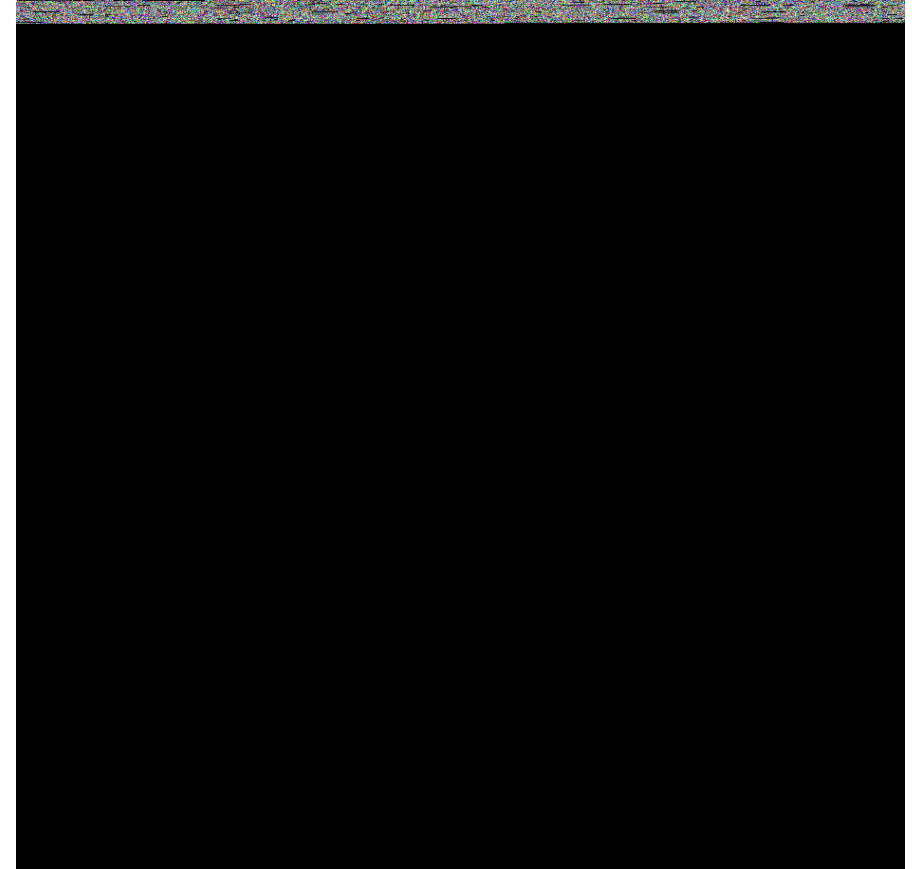
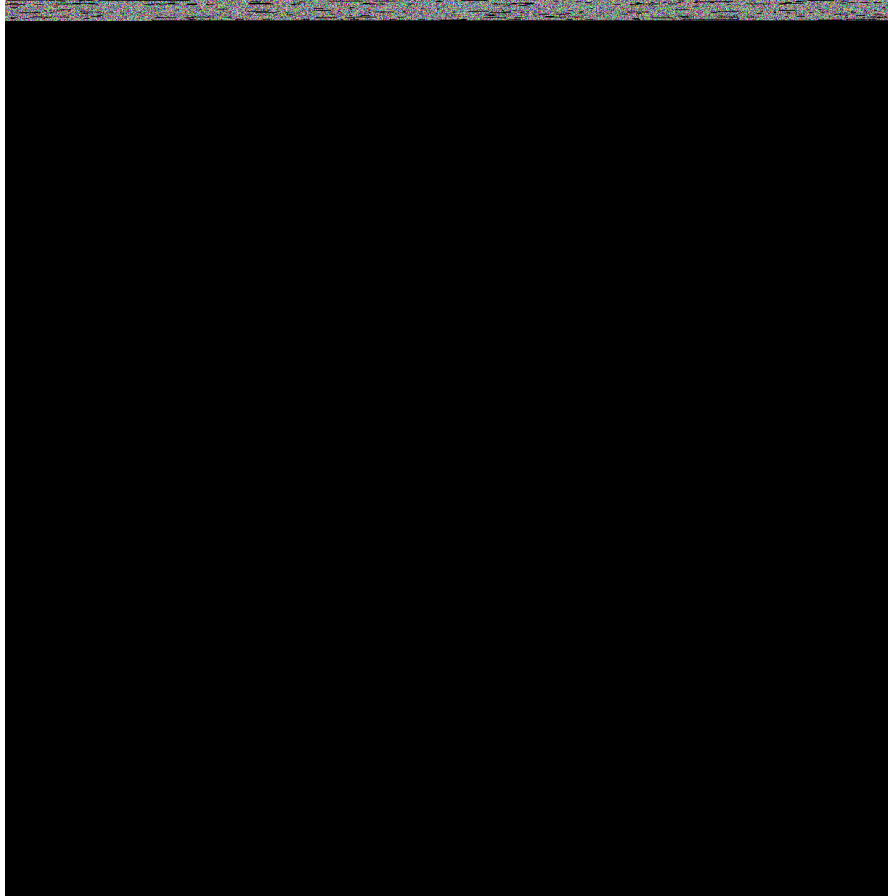


Contribution of Elastic Region



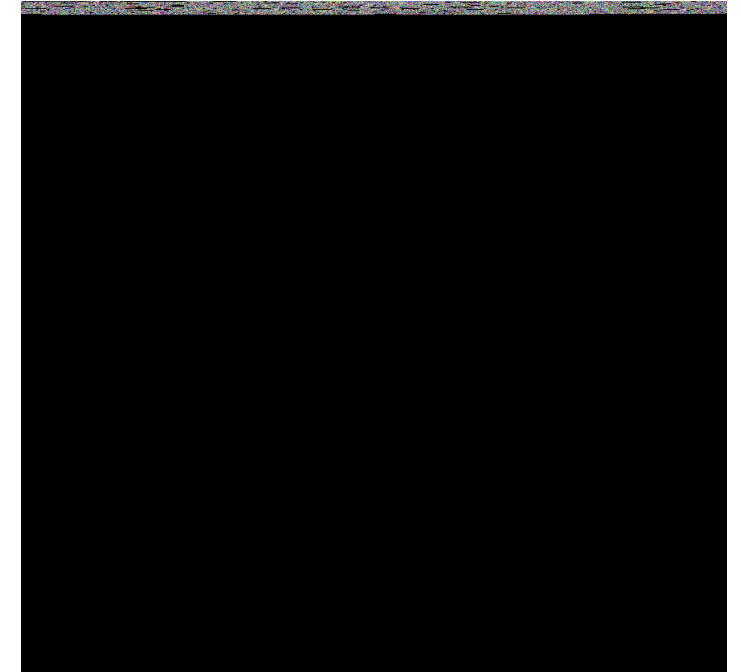
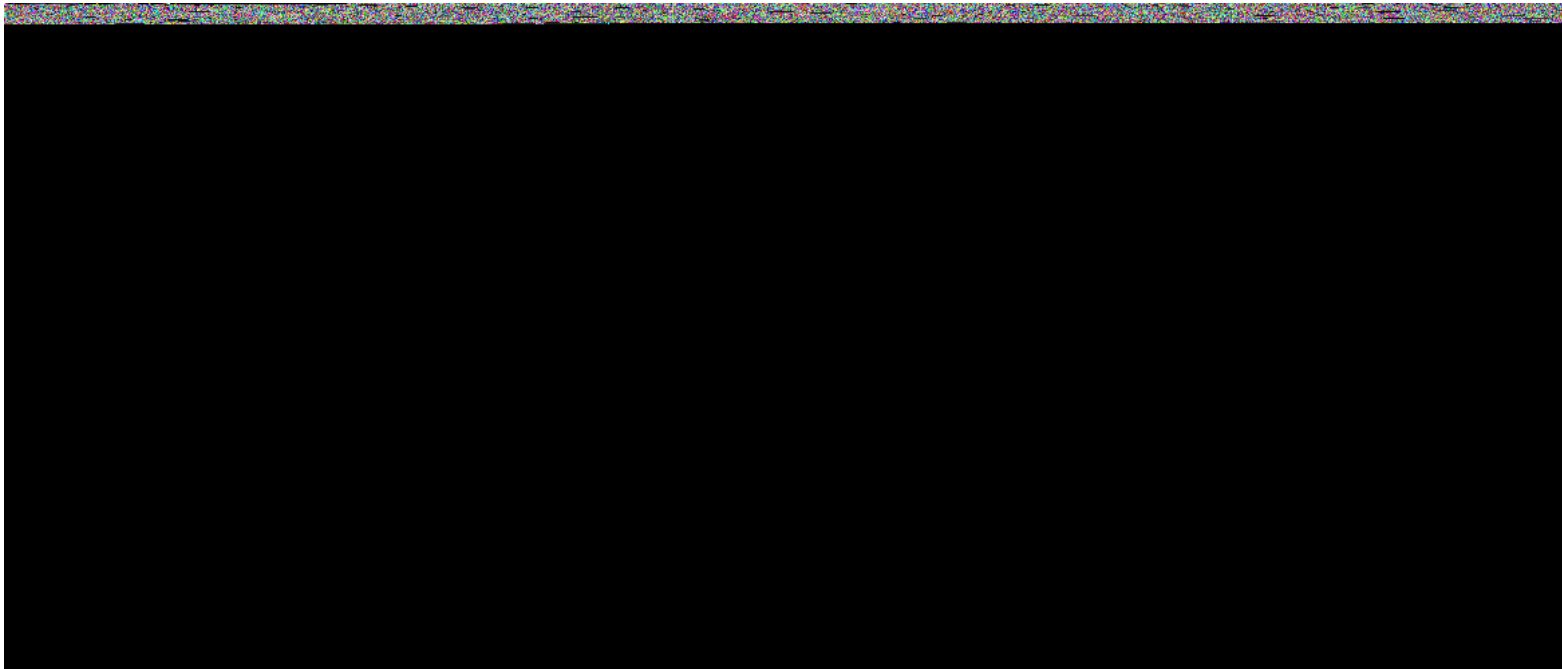
What is the utility of the modulus of rupture as an indicator of performance of fibers?

- Effect of initial tensile cracking



Patel D, Pleesudjai C, Bakhshi M, Nasri V, Mobasher B. Back-calculation of mechanical properties of fiber reinforced concrete in tunnel lining segments. Struct Concr. 2025; (in press). suco.202400914.R2.

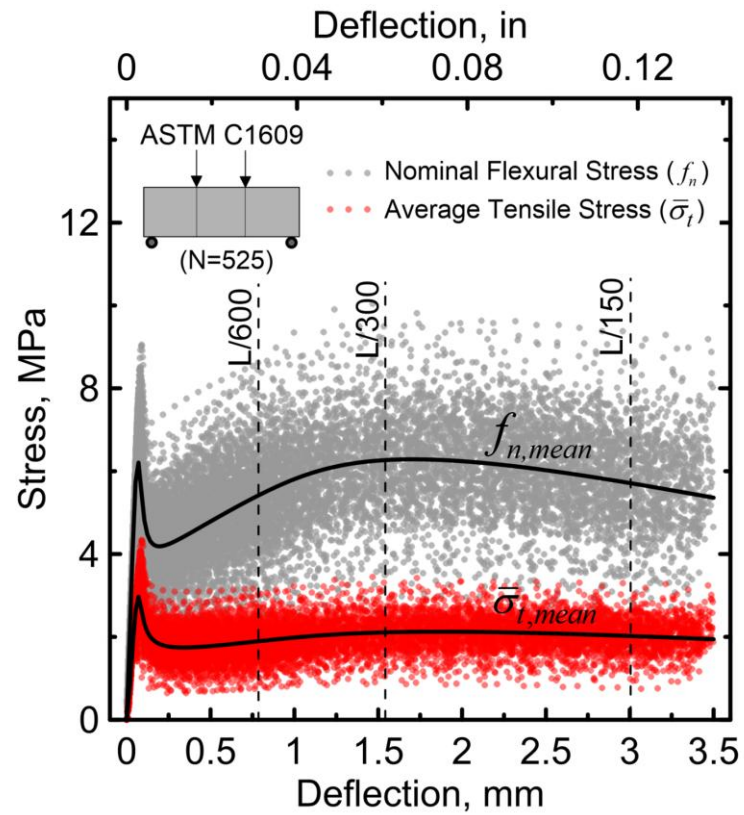
How does the number of samples affect the Characteristic Value?



A sample set of 25 reduces CV within 5% or less.

Use of simplified approach

Characterizing average residual tensile stress:



Patel D, Pleesudjai C, Bakhshi M, Nasri V, Mobasher B. Back-calculation of mechanical properties of fiber reinforced concrete in tunnel lining segments. Concr. 2025; (in press). Manuscript ID: suco.202400914.R2.

Recent Publications

- Yiming Yao, Barzin Mobasher, Jingquan Wang, Qizhi Xu, “Analytical approach for the design of flexural elements made of reinforced ultra-high-performance concrete”, *Structural Concrete*, May 2020 <https://doi.org/10.1002/suco.201900404>
- Yao, Yiming, Karan Aswani, Xinmeng Wang, and Barzin Mobasher. 2018. “Analytical Displacement Solutions for Statically Determinate Beams Based on a Trilinear Moment–Curvature Model.” *Structural Concrete*. <https://doi.org/10.1002/suco.201700150>
- Marangon, Ederli, Barzin Mobasher, Wagner Dambros Fernandes, Flávio de A Silva, Dimas A.S. Rambo, Ronaldo C Battista, Eduardo M.R. Fairbairn, and Romildo D Toledo Filho. 2025. “Experimental Investigation of Tension Stiffening in Hybrid Reinforced Concrete Ties.” *Construction and Building Materials* 458 (January): 139505. <https://doi.org/10.1016/j.conbuildmat.2024.139505>
- Pleesudjai, Chidchanok;, Devansh Deepak; Patel, and Barzin Mobasher. 2023. “Generalized Nonlinear Moment–Curvature Model for Serviceability-Based Design of Hybrid Reinforced Concrete.” *Journal of Structural Engineering* 149 (12). <https://doi.org/10.1061/JSENDH.STENG-12235>
- Patel, Devansh, Chidchanok Pleesudjai, Yiming Yao, Steve Schaef, and Barzin Mobasher. 2022. “Validation Testing of Precast Tunnel Lining Segments Using Polymeric Fibers.” In *RILEM Bookseries*, 36:820–30. Springer Science and Business Media B.V. https://doi.org/10.1007/978-3-030-83719-8_70
- C. Pleesudjai, D. Patel, K. A. Williams Gaona, M. Bakhshi, V. Nasri, and B. Mobasher. 2024. “Statistical Process Control of Fiber-Reinforced Concrete Precast Tunnel Segments.” *ACI Materials Journal* 121 (2): 127–39. <https://doi.org/10.14359/51740373>
- Patel, Devansh, Chidchanok Pleesudjai, Van Bui, Paul Pridemore, Steve Schaef, and Barzin Mobasher. 2023. “Mechanical Response of Precast Tunnel Segments with Steel and Synthetic Macro-Fibers.” *Cement and Concrete Composites* 144 (August): 105303. <https://doi.org/10.1016/j.cemconcomp.2023.105303>
- Pleesudjai, Chidchanok;, Devansh; Patel, Kenneth; Williams, Mehdi; Bakhshi, Varya; Nasri, and Barzin; Mobasher. 2024. “Statistical Process Control of Fiber-Reinforced Concrete Precast Tunnel Segments.” *ACI Materials Journal* 121 (2): 20. <https://doi.org/10.14359/51740373>.
- Pleesudjai, Chidchanok, Devansh Patel, Narayanan Neithalath, and Barzin Mobasher. 2023. “Serviceability-Based Flexural Design of UHPC Beams.” In *Third International Interactive Symposium on Ultra-High Performance Concrete*. Iowa State University Digital Press. <https://doi.org/10.21838/uhpc.16691>.
- Monteiro, Vitor Moreira de Alencar, Daniel Carlos Taissum Cardoso, Flávio de Andrade Silva, and Barzin Mobasher. 2024. “The Influence of Steel Fibers on the Bond Slip Behavior between Rebars and Concrete: Experimental and Analytical Investigation.” *Construction and Building Materials* 411 (January): 134357. <https://doi.org/10.1016/j.conbuildmat.2023.134357>.
- Monteiro, Vitor Moreira de Alencar, Daniel Carlos Taissum Cardoso, Flávio de Andrade Silva, and Barzin Mobasher. 2023. “Closed-Form Solutions for Flexural Fatigue Mechanical Degradation of Steel Fiber Reinforced Concrete Beams.” *Construction and Building Materials* 409 (December): 134200. <https://doi.org/10.1016/j.conbuildmat.2023.134200>.